

04 Data Preparation

Markéta Tranberg, mlap@viauc.dk

February 2026

Table of Contents

1 Why prepare data?

2 Missing data

3 Outliers

4 Scaling

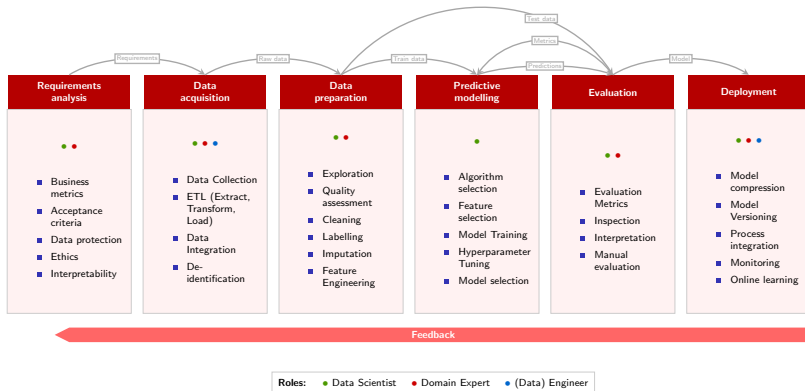
5 String data

6 Feature engineering

7 Assignments

8 Code example

Machine Learning Workflow



Why prepare data?

Garbage in, garbage out.

Raw dataset

Travelling salesperson

ID	Years in Business	Total Sales (\$)	Region	Gender	Avg Discount (%)	Customer Satisfaction	Training Hours
1	2	200 000	North	Male	NaN	3.5	400
2	5	550 000	NaN	Female	NaN	4.0	50
3	10	980 000	West	Male	14.3	NaN	10
4	1	80 000	North	Female	NaN	5.0	100
5	15	1 600 000	North	Male	NaN	4.5	10
6	7	900 000	East	Female	NaN	4.2	5
7	20	2 100 000	South	Male	10.1	2.5	200

What to clean and how

- Irrelevant columns
- Missing values
- Outliers
- Scale
- String data
- Highly correlating features
- Visualization of data distribution
- Imputation
- Removal of rows/columns
- Transformation of data
- Bag of words (count vectorizer)
- One-hot encoding
- Feature selection
- Feature extraction
- Correlation matrix

Irrelevant column

ID	Years in Business	Total Sales (\$)	Region	Gender	Avg Discount (%)	Customer Satisfaction	Training Hours
1	2	200 000	North	Male	NaN	3.5	400
2	5	550 000	NaN	Female	NaN	4.0	50
3	10	980 000	West	Male	14.3	NaN	10
4	1	80 000	North	Female	NaN	5.0	100
5	15	1 600 000	North	Male	NaN	4.5	10
6	7	900 000	East	Female	NaN	4.2	5
7	20	2 100 000	South	Male	10.1	2.5	200

Table of Contents

- 1 Why prepare data?
- 2 Missing data**
- 3 Outliers
- 4 Scaling
- 5 String data
- 6 Feature engineering
- 7 Assignments
- 8 Code example

Missing values

Years in Business	Total Sales (\$)	Region	Gender	Avg Discount (%)	Customer Satisfaction	Training Hours
2	200 000	North	Male	NaN	3.5	400
5	550 000	NaN	Female	NaN	4.0	50
10	980 000	West	Male	14.3	NaN	10
1	80 000	North	Female	NaN	5.0	100
15	1 600 000	North	Male	NaN	4.5	10
7	900 000	East	Female	NaN	4.2	5
20	2 100 000	South	Male	10.1	2.5	200

Handling of missing values

- Remove
 - Delete row
 - Delete column ($\approx > 20\%$)
- Impute
 - Categorical
 - Mode
 - "Missing"
 - kNN (or other classifier)
 - MICE
 - Numerical
 - Mean, median, mode
 - 0
 - kNN/linear regression
 - MICE

```
from sklearn.impute import KNNImputer
```

MICE = Multiple Imputations by Chained Equations

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

A	B	C
	4.2	7.8
3.1	3.1	
4.3		6.3
9.8	5.5	8.1

Reasons for missing data

- **MNAR** (= missing not at random)
Probability of missing X depends on the value of X .
- **MAR** (= missing at random)
Probability of missing X does not depend on the value of X , but may depend on other features.
- **MCAR** (= missing completely at random)
Probability of missing X does not depend on any features at all.

Example: clinical trial for a new blood pressure medication

Fixed missing data

Years in Business	Total Sales (\$)	Region	Gender	Customer Satisfaction	Training Hours
2	200 000	North	Male	3.5	400
5	550 000	North	Female	4.0	50
10	980 000	West	Male	3.95	10
1	80 000	North	Female	5.0	100
15	1 600 000	North	Male	4.5	10
7	900 000	East	Female	4.2	5
20	2 100 000	South	Male	2.5	200

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers**
- 4 Scaling
- 5 String data
- 6 Feature engineering
- 7 Assignments
- 8 Code example

Outliers

Outlier = value outside the normal range

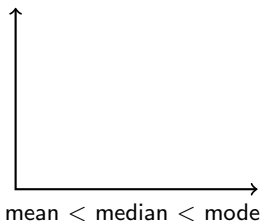
- Keep (if mild)
- Delete
- Impute
- Transform

Boxplot

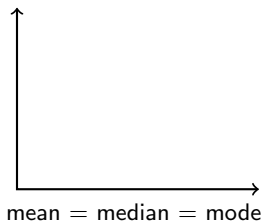
- IQR = interquartile range
- min, max, Q1, median, Q3
- mild/extreme outlier ($> 1.5/3$ IQR from median)

Skewness

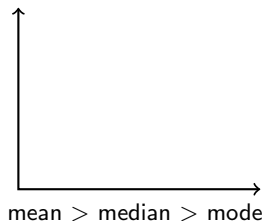
Negative/Left Skew



Normal Distribution



Positive/Right Skew

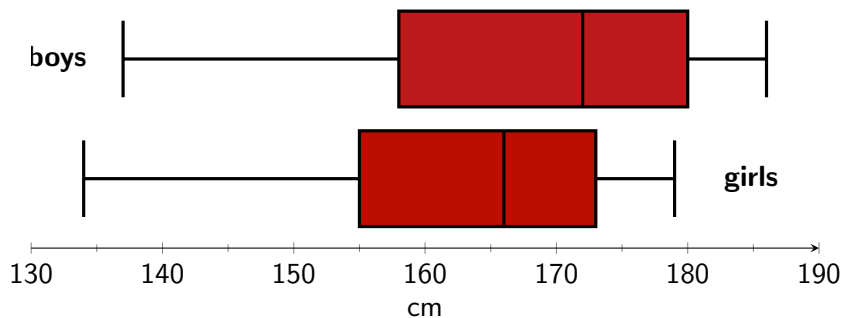


Transformation of skewed data

Left-skewed: x^2 , e^x

Right-skewed: $\log x$, $\sqrt{x}$

Boxplot quiz



Boxplot quiz

- 1 Girls are, on average, taller than boys.
- 2 The total spread (range) of heights is greater for girls than for boys.
- 3 The shortest person in the entire dataset is a girl.
- 4 The tallest person in the entire dataset is a boy.
- 5 Both datasets exhibit a left-skewed distribution.
- 6 The mean height for the boys' dataset is 172 cm.
- 7 Exactly half of the girls measured are between 155 and 173 cm tall.
- 8 The median height of boys is less than the median height of girls.
- 9 Exactly half of the boys are shorter than 172 cm.
- 10 Exactly half of the girls are taller than 173 cm.
- 11 Exactly 3/4 of the girls are taller than 155 cm.
- 12 Exactly 3/4 of the boys are shorter than 180 cm.
- 13 The population displayed in this data is of ethnic Danish descent.
- 14 The Interquartile Range (IQR) for boys is larger than the Interquartile Range for girls.
- 15 At least 25% of the boys are taller than the tallest girl.
- 16 The middle 50% of boys have a height between 158 cm and 180 cm.
- 17 The range of heights for the girls is 45 cm.
- 18 The first quartile (Q1) for boys is higher than the first quartile (Q1) for girls.

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers
- 4 Scaling**
- 5 String data
- 6 Feature engineering
- 7 Assignments
- 8 Code example

Scaling data

Many algorithms using distance metrics are sensitive to scale (kNN, Ridge, Lasso).

Years in Business	Total Sales (\$)	Region	Gender	Customer Satisfaction	log Training Hours
2	200 000	North	Male	3.5	5.99
5	550 000	North	Female	4.0	3.91
10	980 000	West	Male	3.95	2.30
1	80 000	North	Female	5.0	4.61
15	1 600 000	North	Male	4.5	2.30
7	900 000	East	Female	4.2	1.61
20	2 100 000	South	Male	2.5	5.30

Types of scaling

Min-max scaler

- all values between 0 and 1
- $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$
- Preserves relative distances between the values
- Sensitive to outliers

Standard scaler

- $\bar{x}' = 0$ (mean), $\sigma' = 1$ (st.d.)
- $x' = \frac{x - \mu}{\sigma}$
- Re-centers data around 0
- Does not bound data to a specific range (most values $\mu \pm 3\sigma$)
- Assumes data is roughly Gaussian

Scaled data

Years in Business	Total Sales (\$)	Region	Gender	Customer Satisfaction	log Training Hours
-0.89	-1.06	North	Male	-0.61	1.46
-0.45	-0.54	North	Female	0.07	0.13
0.30	0.09	West	Male	0.00	-0.91
-1.04	-1.23	North	Female	1.42	0.57
1.04	1.01	North	Male	0.75	-0.91
-0.74	-0.02	East	Female	0.34	-1.35
1.79	1.74	South	Male	-1.97	1.02

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers
- 4 Scaling
- 5 String data**
- 6 Feature engineering
- 7 Assignments
- 8 Code example

String data

Years in Business	Total Sales (\$)	Region	Gender	Customer Satisfaction	log Training Hours
-0.89	-1.06	North	Male	-0.61	1.46
-0.45	-0.54	North	Female	0.07	0.13
0.30	0.09	West	Male	0.00	-0.91
-1.04	-1.23	North	Female	1.42	0.57
1.04	1.01	North	Male	0.75	-0.91
-0.74	-0.02	East	Female	0.34	-1.35
1.79	1.74	South	Male	-1.97	1.02

String representation

- **Text**
Bag of words
- **Category**
One-hot encoding

Bag of Words

Sentence 1:

The green frog jumps over the small frog.

Sentence 2:

A quick lizard runs past the green lizard.

Note: Usually, we remove punctuation and convert to lowercase during preprocessing.

```
from sklearn.feature_extraction.text import  
CountVectorizer
```

a
frog
green
jumps
lizard
over
past
quick
runs
small
the

One-Hot Encoding

Categorical



One-Hot Encoded

Region
North
North
West
North
North
East
South

North	West	East	South

Note: Binary classification can be expressed with one column.

Handled string values

Years in Business	Total Sales (\$)	Reg. North	Reg. West	Reg. East	Reg. South	Gen. Male	Cust. Sat.	log T.H
-0.89	-1.06	1	0	0	0	1	-0.61	1.46
-0.45	-0.54	1	0	0	0	0	0.07	0.13
0.30	0.09	0	1	0	0	1	0.00	-0.91
-1.04	-1.23	1	0	0	0	0	1.42	0.57
1.04	1.01	1	0	0	0	1	0.75	-0.91
-0.74	-0.02	0	0	1	0	0	0.34	-1.35
1.79	1.74	0	0	0	1	1	-1.97	1.02

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers
- 4 Scaling
- 5 String data
- 6 Feature engineering**
- 7 Assignments
- 8 Code example

Feature engineering

- **Feature selection:** selecting relevant features
- **Feature extraction:** combining existing features

Covariance, Correlation

Covariance

Measures the direction of the linear relationship (scale-dependent).

$$\text{cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Pearson Correlation (r)

The standardized version of covariance (dimensionless)

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Note: Remember r^2 ?

Correlation

- **Values:** range from -1 to $+1$.
- **High Positive** ($r \approx 1$): As X increases, Y increases consistently.
- **High Negative** ($r \approx -1$): As X increases, Y decreases consistently.
- **Magnitude:** $|r| > 0.7$ is generally considered "high".

Correlation matrix

In code :)

Two highly correlating features

Highly correlated features are unwanted.

- Remove one
- Combine them into a new feature

Final data frame

Years in Business	Sales /Years	Reg. North	Reg. West	Reg. East	Reg. South	Gen. Male	Cust. Sat.	log T.H
-0.89	-0.30	1	0	0	0	1	-0.61	1.46
-0.45	0.44	1	0	0	0	0	0.07	0.13
0.30	-0.45	0	1	0	0	1	0.00	-0.91
-1.04	-1.78	1	0	0	0	0	1.42	0.57
1.04	0.20	1	0	0	0	1	0.75	-0.91
-0.74	1.82	0	0	1	0	0	0.34	-1.35
1.79	0.07	0	0	0	1	1	-1.97	1.02

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers
- 4 Scaling
- 5 String data
- 6 Feature engineering
- 7 Assignments**
- 8 Code example

Assignments

- 1 Car Prices - done
- 2 Candidates I
- 3 Mushrooms
- 4 Exoplanets
- 5 Sentiment analysis
- 6 Candidates II

All include at least some pre-processing

Table of Contents

- 1 Why prepare data?
- 2 Missing data
- 3 Outliers
- 4 Scaling
- 5 String data
- 6 Feature engineering
- 7 Assignments
- 8 Code example**

Code example

Github link: https://github.com/304125/MAL1_S26/tree/master/04_Data_Preparation